TECHNICAL DOCUMENTATION

BetonWin Support Widget

AI-Powered Customer Support & Zendesk Integration

Version 4.0 · March 2026

Architecture, Features, n8n Workflows & Zendesk Classification

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1 Project Overview

BetonWin Support Widget is an AI-powered customer support chat widget embedded on the **beton.win** gambling platform. It provides 24/7 automated support in 4 languages, handles deposit verification flows, and seamlessly escalates to human agents via Zendesk when needed.

Key Facts:

· Single JavaScript file (`widget.js`, ~1,550 lines) — all CSS inline
· Embedded on GR8/Modulor platform (beton.win)
· 10 specialized AI agents on n8n for different support topics
· Zendesk integration with automatic ticket classification (L1/L2/L3)
· Languages: Spanish (default), English, Italian, Portuguese

2 Architecture & Hosting

System Architecture

COMPONENT	TECHNOLOGY	DETAILS
Widget	Vanilla JS	Single file, all CSS inline, self-contained IIFE
AI Backend	n8n + OpenRouter	10 specialized agents with Pinecone vector stores
Knowledge Base	Google Apps Script + Pinecone	Searchable KB via GET request
Ticketing	Zendesk	Ticket creation, agent polling, comment sync
File Storage	AWS S3	Presigned URL upload for payment proofs

Hosting & Deployment

ITEM	URL / DETAILS
GitHub Repo	`stepbystepautomatization-jpg/betonwin-widget`
CDN (jsDelivr)	`cdn.jsdelivr.net/gh/stepbystepautomatization-jpg/betonwin-widget@main/widget.js`
Cache Purge	`purge.jsdelivr.net/gh/.../widget.js`
n8n Instance	`https://n8nbeton.online`
Zendesk	`betonwin.zendesk.com`

Deployment Process: Push to GitHub → Purge jsDelivr cache → Widget updates automatically on beton.win

3 Widget Features

Core Chat

- AI chat powered by 10 specialized n8n agents (Deposits, Withdrawals, Bonuses, Verification, Account, Technical, Sports, Casino, Balance, General)
- Knowledge base search via Google Apps Script + Pinecone vector stores
- Auto-detect language from user text (ES, EN, IT, PT)
- Markdown rendering (bold, italic, lists, links, horizontal rules)
- Rate limiting (1s cooldown, max 1000 chars per message)
- Safety timer: auto-recovers if stuck for 30 seconds

Player Detection

- Auto-detect Player ID, username, and email from the host site
- Methods: localStorage scan, `window.__USER__`, DOM elements, cookies
- Session UUID (v4) generated on init, attached to all API calls for analytics correlation

Deposit Verification Flow

- Player ID input → deposit status check via n8n
- File upload via S3 presigned URL (JPG, PNG, PDF — max 10MB)
- AI document analysis with async polling
- Results: APPROVED, REJECTED, PENDING_REVIEW, TIMEOUT

UI Features

FEATURE	DESCRIPTION
Emoji Picker	24 contextual emojis (gambling, payments, feedback) — button in input area
Fullscreen Mode	Expand button in header, toggles full viewport mode
Smart Typing Indicator	"BetonWin AI thinking..." for bot, "María thinking..." for real agent
Notification Badge	Unread message counter + notification sound
Proactive Message	Badge notification after 30s if user hasn't interacted
CSAT Rating	1-5 star rating shown after agent closes conversation
Quick Actions	"Check deposit status" button for common flows

Design

Theme: Dark teal glass (`#03242D`) with green accents (`#2DEC76`)
Font: Inter (loaded async via preload — non-render-blocking)

Responsive: 400×620px desktop, fullscreen on mobile (<440px)

Bot Bubble: Dark with green border · **User Bubble:** Green gradient · **Agent Bubble:** Amber accent

4 Chat Flow & State Machine

Main Chat Flow

CHAT → DEPOSIT_ASK_ID → DEPOSIT_CHECKING → UPLOAD_REQUIRED → UPLOADING → ANALYZING → RESULT

Escalation Flow

CHAT → ESCALATING → LIVE_AGENT → AGENT_CLOSED

Message Processing Pipeline

1. **Legal threat check** — local keyword scan → immediate escalation if detected
2. **Deposit intent check** — keyword detection → starts deposit verification flow
3. **KB search** — Google Apps Script query (1.5s timeout, race with empty result)
4. **AI call** — send message + KB content + history to n8n (12s timeout)
5. **Response analysis** — check for language rejection → retry in Spanish if needed
6. **Human request detection** — analyze AI response for acknowledgment signals
7. **Display response** — render with markdown parsing

5 AI Escalation System

The escalation system uses a **3-tier approach** — no rigid keyword matching for human requests.

TIER	TRIGGER	ACTION	SPEED
LEGAL	Local keyword scan (avvocato, denuncia, lawsuit, attorney, etc.)	Immediate escalation to Zendesk	Instant
HUMAN REQUEST	AI response analysis — detects when AI acknowledges user wants a human	Count +1. After 3 detections → escalation	After AI responds
NORMAL	No escalation signals detected	Show AI response normally	—

How AI-Based Detection Works

1. Every message goes to the AI first (except legal threats)
2. After the AI responds, the widget scans the response for **acknowledgment signals**:
   - "entiendo que prefieres hablar con una persona"
   - "capisco che preferiresti parlare con un agente"
   - "I understand you'd prefer to speak with someone"
   - "frustración", "non posso metterti in contatto", etc.

3. If the AI acknowledges the human request → counter increments
4. At count 3 → automatic escalation to Zendesk with full conversation history
5. Counter resets on escalation failure to prevent infinite loops

> **Why AI-based?** Works with any language, any phrasing. "Basta col bot", "pasame alguien de verdad", "this is useless let me talk to someone" — the AI understands the intent and the widget reads the AI's acknowledgment.

6 Live Agent (Zendesk) Integration

Escalation Process

1. Widget sends conversation history + player info to `/webhook/escalate`
2. n8n classifies the conversation → determines Topic, L1, L2, L3, Priority
3. n8n assigns the correct **Zendesk Form** based on category
4. Zendesk ticket created with HTML body (light theme, readable on any Zendesk theme)
5. Widget enters LIVE_AGENT mode — starts polling for agent replies every 5 seconds

Agent Communication

FEATURE	IMPLEMENTATION
Real agent name	Zendesk API `?include=users` sideload → resolves `author_id` to name
Agent name in header	Header updates to show "María" instead of generic "Live Agent"
Typing indicator	"María thinking..." shown with agent's real name
File sharing	User can upload files → S3 → URL sent as Zendesk comment
Widget close notification	If user closes widget during live session, agent is notified on Zendesk
CSAT rating	1-5 star rating shown after conversation closes, sent to Zendesk

Comment Filtering

The polling system filters comments to show only agent replies:

- Skip messages containing `[Widget User]` (sent by the widget via n8n)
- Skip messages containing `Bot Conversation Transcript` (initial ticket body)
- Only show comments newer than last poll timestamp

> **Important:** Chat only closes on Zendesk status `closed` — NOT on `solved` or `pending`. This prevents premature closure when Zendesk auto-sets status on agent reply.

Ticket HTML Structure

Every ticket includes a formatted HTML body with explicit colors (white background, dark text) for readability on any Zendesk theme:

- **Header:** Player ID, Username, Email, Language, Escalation Reason, Category
- **Quick Recap Box:** Yellow background, customer needs summary, last message, message count
- **Conversation Transcript:** User messages (blue background), Bot messages (gray background), with timestamps

7 n8n Workflows

⚠️ **DO NOT modify the "widget cs" workflow** — it is unrelated to the widget and should never be touched.

Workflow 1: BetonWin AI Support v3.0 — 10 Agents

PROPERTY	VALUE
Workflow ID	`CfS-ypkrpmP5fq-SB1qKN`
Webhook	`/webhook/884acc5b-3a84-44cc-8f44-7c1b7df3df2a`
Status	**Active**

Node Pipeline

Webhook → **Extract Query** → **Router** → **Category Switch** → **Agent (x10)** → **Format Response** → **Respond**

10 Specialized Agents

AGENT	SCOPE	KB (PINECONE)
Deposits	Deposit methods, amounts, timing, rejected deposits, refunds	KB Deposits
Withdrawals	How to withdraw, limits, rejected withdrawals, 70% rule	KB Withdrawals
Bonuses	Welcome bonus, rollover, free spins, cashback, promo codes	KB Bonuses
Verification	KYC, documents, email/phone verification	KB Verification
Account	Registration, password, blocked accounts, close account	KB Account
Technical	Site errors, game issues, login problems, performance	KB Technical
Sports	Betting types, odds, parlays, live betting, cash out	KB Sports
Casino	Slots, roulette, blackjack, live casino, RTP	KB Casino
Balance	Total vs available balance, locked funds, cashback	KB Balance
General	Platform info, responsible gambling, support channels	KB General

Format Response Node

Strips `<!--lang:XX-->` tags, extracts `<!--action:XXX-->` tags, returns clean JSON with optional `action` field.

Workflow 2: BetonBot — Zendesk Escalation

PROPERTY	VALUE
Workflow ID	`QUHhAtOeh6lFb9XE`
Status	**Active**

Endpoints

ENDPOINT	METHOD	PURPOSE
`/webhook/escalate`	POST	Create Zendesk ticket with conversation history & auto-classification
`/webhook/agent-reply`	POST	Add user comment to existing ticket
`/webhook/agent-poll`	POST	Poll for new agent replies (with `?include=users` for real names)

Format Ticket Node

Classifies conversation into Zendesk categories using keyword scoring, then assigns the correct Form, Topic, Type of Contact, Priority, and L1/L2/L3 custom fields.

8 Zendesk Custom Fields — Full Classification

Zendesk Forms

FORM NAME	FORM ID	USED WHEN
General	29325705286290	General questions, registration, refunds
Account	29328404236050	Account management, password, close account
Issues	29328455838866	Technical problems, login, game crashes
Verification	29328593704466	KYC, email/phone verification
Product	29328700357394	Casino, sports, slots info
Promotions and bonus	29328754437266	Bonuses, promo codes, loyalty program
Withdrawal	29342990784274	Withdrawals, limits, declined
Deposits	29343128324242	Deposits, payment methods, declined
VIP Topics	29344679287826	VIP perks, birthday, follow-up
Ghost Contact	29351149783058	No real contact / empty messages

Type of Contact

VALUE	WHEN USED
`question`	Informational questions
`request`	Action requests (close account, VIP, etc.)
`non_technical_issue`	Non-technical problems (missing deposit, withdrawal)
`technical_issue`	Technical problems (crashes, errors, login)
`lack_of_customer_knowledge`	Customer doesn't know how something works
`ghost_contact2`	Ghost / empty contact
`follow_up`	Follow-up on previous request

Priority Mapping

PRIORITY	WHEN APPLIED
Urgent	Legal threats, account restrictions, duplicated accounts, verification declined/unable, technical login/website errors
High	Deposit not reflected, deposit delay, deposit approved/pending, account close (gambling), VIP birthday, VIP missed call

Medium	Deposit declined, withdrawal declined, withdrawal limit, account changes, password reset, verification
Low	Promotions, product info, general questions, how-to queries

DEPOSITS Classification

L1	L2	L3	PRIORITY	ZENDESK VALUE
How to deposit	—	—	Medium	`how_to_deposit`
Payment methods offered	—	—	Low	`payment_methods_offered`
Deposit status	Approved	—	High	`deposit_status → approved4`
Deposit status	Pending	—	High	`deposit_status → pending4`
Deposit status	Declined	Timeout	Medium	`deposit_status → declined4 → timeout`
Deposit status	Declined	Declined by bank	Medium	`→ declined_by_bank`
Deposit status	Declined	General declined	Medium	`→ general_declined`
Deposit status	Declined	Antifraud declined	Medium	`→ antifraud_declined`
Deposit status	Declined	Too many requests	Medium	`→ too_many_requests`
Deposit not reflected	—	—	High	`deposit_not_reflected`
Deposit Delay	—	—	High	`deposit_delay`
General info	—	—	Low	`general_info___`

WITHDRAWAL Classification

L1	L2	L3	PRIORITY	ZENDESK VALUE
How to withdrawal	—	—	Low	`how_to_withdrawal`
Withdrawal Status	Pending	—	Low	`withdrawal_satus → pending12`
Withdrawal Status	Declined	Timeout	Medium	`→ declined12 → timeout2`
Withdrawal Status	Declined	Declined by bank	Medium	`→ declined_by_bank2`
Withdrawal Status	Declined	General declined	Medium	`→ general_declined2`
Withdrawal Status	Declined	Antifraud declined	Medium	`→ antifraud_declined2`
Withdrawal Status	Declined	Too many requests	Medium	`→ too_many_requests2`
Withdrawal Status	Declined	70% not wagered	Low	`→ 70__not_wagered1`
Withdrawal Status	Approved	—	Low	`→ approved12`
Unable to Withdraw	Incomplete verif.	—	Low	`unable_to_withdraw`
Cancel Request	—	—	Low	`cancel_request2`

Withdrawal limit	—	—	Medium	`withdrawal_limit_`

ACCOUNT Classification

L1	L2	PRIORITY	ZENDESK VALUE
Request to close account	Gambling addiction	High	`request_to_close_account → gambling_addiction1`
Request to close account	Self exclude	Urgent	`→ request_to_self_exclude_`
Request to close account	Incorrect country	Low	`→ incorrect_country/currency`
Duplicated account	—	Urgent	`duplicated_account_`
Change personal details	—	Medium	`change_personal_details`
Update phone number	—	Medium	`update_phone_number`
Update email	—	Medium	`update_email`
Balance info	—	Low	`balance_info`
Password Reset	—	Medium	`password_reset`
Account restrictions	—	Urgent	`account_restrictions`
Delete data request	—	Medium	`delete_data_request`
Email not found	—	Low	`email_not_found`
Re-open account	—	Medium	`request_to_re-open_account_`

ISSUES Classification (Technical / Non-Technical)

L1	L2	PRIORITY	ZENDESK VALUE
Technical	Login Problems	Urgent	`technical → troubleshooting`
Technical	Website/Platform Errors	Urgent	`→ blocked_game1`
Technical	Bonus Not Triggering	Urgent	`→ frozen_page1`
Technical	Game Not Loading/Crashing	Low	`→ client_cant_log_in1`
Technical	Deposit Payment Error	Urgent	`→ deposit_payment_error__tech_`
Technical	Withdrawal Payment Error	Urgent	`→ withdrawal_payment_error__tech_`
Technical	Performance/Slow	Low	`→ unknown_error1`
Technical	Error in Game/Bonus	Low	`→ other11`
Technical	Incorrect Balance	Medium	`→ incorrect_balance_/_funds_not_updated`
Technical	UI Issues	Low	`→ ui_issues`

Technical	Bet not settled	Medium	→ `bet_not_settled_or_settled_incorrectly_`
Non Technical	BTS	Low	`verification2` → `bts`
Non Technical	Incorrect bonus offer	Low	→ `incorrect_bonus_offer_sent`

PROMOTIONS Classification

L1	L2	PRIORITY	ZENDESK VALUE
Retracted Bonus	Expired Bonus	Low	`retracted_bonus → expired_bonus2`
Retracted Bonus	Max Limit bonus winning	Low	`→ max_limit_bonus_winning2`
Withdrawal unavailable by Bonus Terms	—	Low	`free_spins_bonus`
Balance locked	—	Low	`deposit_bonus__`
Cancel Request	—	Low	`free_bet_bonus`
General info	Terms and conditions	Low	`cashback_bonus → terms_and_conditions3`
General info	Other bonus info	Low	`→ oher_bonus_info`
Loyalty Program Info	—	Low	`daily_bonus_info`
Bonus Request	—	Low	`bonus_request2`
Sport Bonus	Mechanic (50/50)	Low	`sport_bonus_ → bonus_mechanic__50/50_...`
Sport Bonus	Terms and conditions	Low	`→ terms_and_conditions11`
Promo code	—	Low	`promo_code_`

VERIFICATION Classification

L1	L2	PRIORITY	ZENDESK VALUE
Email verification	—	Medium	`email_verification`
Phone verification	—	Medium	`phone_verification`
How to verify	—	Medium	`how_to_verify`
Verification status	Completed	Medium	`verification3 → completed1`
Verification status	Declined	Urgent	`verification3 → declined1`
User unable to verify	—	Urgent	`user_unable_to_verify_account`
Risk Verification	—	Urgent	`risk_verification`

PRODUCT, VIP & GENERAL

CATEGORY	L1 VALUES	PRIORITY
Product	sportbet_general_info, casino_general_info, slot_general_info, live_casino_info, games_info, sportbet_not_available (Medium), product_suggestion,	Low

	sportsbook_general_info	
VIP Perks	VIP Exclusive Promo (Medium), Anniversary/Birthday (High), VIP Bonus Request (Medium)	Medium–High
VIP Follow Up	Missed Call	High
General	general_questions, how_to_register, how_to_play, refund_request, more_info, terms_and_conditions, privacy_policy, navigation_help, fairplay_rng, collaboration	Low

9 API Contracts

POST /webhook/884acc5b-... (AI Chat)

Request

FIELD	TYPE	DESCRIPTION
`message`	string	User's message
`kb_content`	string	Knowledge base content + language instruction
`lang`	string	Detected language (es/en/it/pt)
`history`	array	Last 6 messages [{role, content}]
`session_id`	string	UUID v4 for session correlation

Response

FIELD	TYPE	DESCRIPTION
`response`	string	AI response text
`action`	string|null	WANTS_HUMAN, START_DEPOSIT_FLOW, or null
`sessionId`	string	Session identifier

POST /webhook/escalate

Request

FIELD	TYPE	DESCRIPTION
`reason`	string	"human_request" or "legal_threat"
`language`	string	User's language
`conversation_history`	array	[{role, content, ts}] — must be array, not string
`player_id`	string	Player identifier
`player_name`	string|null	Auto-detected username
`player_email`	string|null	Auto-detected email

Response

FIELD	TYPE	DESCRIPTION
`ticket_id`	number	Zendesk ticket ID

`status`	string	"ESCALATED" or "ERROR"

POST /webhook/agent-poll

Request:

```
{ ticket_id, since }
```

Response:

```
{ comments: [{author, body, ts}], status: "new|open|pending|solved|closed" }
```

POST /webhook/agent-reply

Request:

```
{ ticket_id, message }
```

Response:

```
{ status: "ok" }
```

10 Bug Fixes Applied (17)

#	SEVERITY	FIX
1	High	XSS via URL in markdown links — escape quotes in href
2	High	Crash if `#__beton_widget__` missing — auto-create container
3	Medium	sendFileToAgent missing file type validation
4	Medium	Progress bar never reset after upload
5	Medium	Race condition on escalation — input stays disabled
6	Medium	Retry AI payload missing language instruction
7	Medium	TIMEOUT status not handled in handleFinalResult
8	Medium	Agent polling not stopped on page unload
9	Medium	Agent messages sent as 'user' in AI history
10	Low	@import font render-blocking → async preload
11	Low	Proactive notification after user already interacted
12	Low	showTyping before addMessage in deposit flow
13	Low	File input not reset in resetChat

14	Low	Upload area clickable outside UPLOAD phase
15	Low	addMessage('agent') renders without avatar
16	Low	Missing aria-live for screen reader accessibility
17	Low	iOS audio unlock on first tap